

- 6** Use the substitution $y = vx$ to find the solution of the differential equation

$$x \frac{dy}{dx} = y + \sqrt{9x^2 + y^2}$$

for which $y = 0$ when $x = 1$. Give your answer in the form $y = f(x)$, where $f(x)$ is a polynomial in x .
[10]

[illegible]